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Project Goal:

The goal of this project is to investigate how different temperature treatments influence forage productivity during the establishment of alfalfa and orchardgrass at Pennsylvania State University.

Research Questions:

1. How does the effect of temperature treatment on forage yield vary between years with different environmental conditions?

Response variable: Forage yield (continuous)
Predictor variable: temperature treatment (categorical), year (categorical)

2. How does the effect of temperature treatment on forage productivity differ between weed and weed free plots?

Response variable: Forage yield (continuous)
Predictor variable: Temperature treatment (categorical), weeds (categorical)

3. How does the temperature treatment affect forage productivity at different points in the growing season?

Response variable: Forage yield (continuous)
Predictor variable: Temperature treatment (categorical), harvesting period (categorical)

4. How is forage productivity associated with average temperature during the growing season?

Response variable: Forage yield (continuous)
Predictor variable: Average temperature (continuous)

5. How is forage productivity associated with precipitation during the growing season?

Response variable: Forage yield (continuous)
Predictor variable: Precipitation (continuous)

Data source and description:

The data for this project comes from the USDA Ag Data Commons dataset, "Effect of increasing temperature and temperature fluctuations on alfalfa and orchardgrass in the establishment year." This data was collected at Penn State University during an experiment that was first conducted in 2020-2021 and repeated in 2021-2022. The experiment studied the effects of increased temperature on forage productivity, winter injury, and weed suppression. The goal of this research is to find alternative forages that are more tolerant of heat and dryness and to develop management strategies that improve resilience.

This is important to study because alfalfa and orchardgrass both provide a lot of benefits to the land and the environment. Alfalfa provides nitrogen to other crops, reduces erosion, and provides soil health benefits (USDA). Orchardgrass is often used as grazing material for livestock and is also beneficial for the soil (NRCS). Analysis of this data can help farmers keep their crops and soil healthy while facing rising temperatures and increasing dryness. It is also worth mentioning that this is important to study because global temperatures are rising, so it is imperative to closely monitor agricultural output to proactively look for solutions to future problems.